

Probabilistic Modeling of the Logical Coherence of LLM Responses

Two-Layer Relations, Markov Networks vs. Markov Logic, and Coherence Scoring — a Deep Dive

FactReasoner — Ideation

July 10, 2026

Abstract

We deepen the research plan for measuring the *logical coherence* of an LLM-generated response (as opposed to its factuality). The response is decomposed into atomic claims; *inter-claim* relations are mined and assigned confidences; a probabilistic model over claims and relations is built; and a single coherence score is read out by inference. This document (i) develops the two-layer relation taxonomy (discourse sense $\rightarrow$ inferential coupling); (ii) works the two most promising models — the undirected pairwise **Markov Random Field** (the FactReasoner construction, reused) and the **Markov Logic Network** — rigorously and side by side; (iii) analyzes all applicable scoring methods and computes them *exactly* on a new, deliberately intricate worked example that combines a plausible causal chain, an *unresolved* contradiction, and a *resolved* (conceded) tension. A key modeling finding: for atom $\leftrightarrow$ atom coherence the *with-priors* factor variant already present in `assessor.py` is the correct choice, and a reified coherence node plus the partition-function ratio give the cleanest global scores.

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1 Setup and notation

A response y is decomposed (reusing FactReasoner’s atomizer + decontextualizing reviser) into atomic units $A = \{a_1, \dots, a_n\}$, each a binary random variable: $a_i = 1$ ("holds") or $a_i = 0$. We mine a set of directed inter-atom relations $R = \{r_{ij} = (a_i, a_j, \tau_{ij}, p_{ij})\}$, where τ_{ij} is the relation type and $p_{ij} \in [0, 1]$ its confidence/strength (Section 4). The goal is a *Logical Coherence Score* $LCS(y) \in [0, 1]$ plus per-atom / per-relation diagnostics. This reuses the objects in `src/fact_reasoner/`: `FactGraph` (Node/Edge, where `Edge.link="atom_atom"` is already permitted), `MarkovNetwork` (UAI serialization), and `Merlin` for inference.

Two failure modes must be captured, both drawn from `docs/ideation/*.pdf`: **(1) weak inferential links** (true atoms, tenuous asserted relations) and **(2) internal contradiction / position invalidation** (a later atom retracts or conflicts with an earlier one). Crucially, coherence is *order-sensitive*: the *R v Renda* example (`example-5`) shows the same facts can be coherent or not depending only on arrangement.

2 The two-layer relation taxonomy

We separate what the classifier is good at naming (a *discourse* sense) from what the probabilistic model needs (an *inferential coupling*).

2.1 Layer B — discourse sense (interpretable label)

Grounded in the **PDTB 3.0** four top-level senses [PDTB3, PDTB2] and **RST** [RST]; **SDRT** [SDRT] supplies the graph (not tree) view and a formal "maximize coherence" notion. Layer B is what shallow discourse parsers (**DiscoPrompt** [DiscoPrompt], connective-prediction) actually output.

PDTB top	Sub-senses used here	Dir.?	Typical Layer-A map
Contingency	Cause-Effect, Effect-Cause, Evidence, Condition	yes	ENTAIL (graded)
Temporal	Precedence, Succession, Synchrony	yes	weak/NONE + order constraint
Comparison	Contrast, Concession	yes	CONTRADICT; Concession = <i>resolved</i> tension
Expansion	Elaboration, Restatement, Instantiation/Subsumption	mixed	ENTAIL/EQUIV

2.2 Layer A — inferential coupling (drives the model)

A small set matching FactReasoner’s existing NLI edge types, so the factor machinery is reused verbatim:

- ENTAIL ($a_i \Rightarrow a_j$): source true makes target true (prerequisite, cause-effect, evidence, subsumption).
- CONTRADICT ($a_i \Rightarrow \neg a_j$): source true makes target false (contrast, position invalidation).
- EQUIV: the two agree (restatement, mutual entailment).
- NONE: no coupling (no edge).

The two relations used throughout the ideation PDFs — *logical prerequisite* ($A \rightarrow B$) and *position invalidation* ($A \neq B$) — are exactly ENTAIL and CONTRADICT.

2.3 Why two layers, and three subtleties

1. **The map $B \rightarrow A$ carries a prior on strength.** "Cause-Effect" maps to ENTAIL but with a *graded* p reflecting causal plausibility; "Restatement" maps to EQUIV with high p . We learn/prompt this map.
2. **Direction and asymmetry.** Cause-Effect vs. Effect-Cause and Precedence vs. Succession are directional; we store an ordered pair. Order-sensitivity is what distinguishes *Renda* S from K.
3. **Concession = resolved tension.** A Concession, or a *holding* that adjudicates a dispute, is *not* a coherence defect — it is a contradiction the text itself resolves. It must be *discounted*: modeled as a CONTRADICT edge with reduced p (MRF), or cancelled by a rule when a resolving atom is present (MLN, Section 6). This is the S-vs-K lesson.
4. **Type posterior $\times$ conditional strength.** Two distinct quantities: $P(\tau_{ij} = \text{contradict})$ (is it a contradiction at all?) and $P(a_j \mid a_i, \tau_{ij})$ (given it is causal, how strong is the link?). The model should carry their product; Section 4.

3 A richer worked example: the *AeroParts* recall report

The ideation PDFs isolate one phenomenon each (pure prerequisite; a single pivot invalidation; a reordering effect). We now combine *all* of them in one passage, so the models and scores visibly diverge. The passage has a causal/evidential backbone, one **unresolved** factual contradiction (two irreconcilable casualty counts), one **resolved** tension (an early blame claim retracted by an adjudicated holding — a concession), and one **equivalence** (a restatement).

Source paragraph

Plain text.

AeroParts shipped a batch of turbine blades certified to spec, and the blades were installed in the carrier's A-320 fleet. The blades then entered regular service across the fleet. During service, seventeen blades failed in flight. The regulator opened a safety probe, and within a week issued an order grounding the fleet; in effect, every affected aircraft was pulled from service. Maintenance logs, later disclosed, recorded abnormal vibration on the failed units. The airline stated that no passengers or crew were harmed in any of the incidents. A separate wire report, however, claimed that three passengers died in one of the failures. Early in the inquiry the airline argued that the failures were caused by pilot error. The final accident report concluded that the root cause was a metallurgical defect in the AeroParts blades, and on that basis found the pilots not at fault. Because AeroParts had shipped the defective batch, the tribunal named AeroParts liable and ordered it to pay damages. The regulator also published a preliminary bulletin.

Annotated. Each clause is one atomic unit ^{an}. Colours mark roles: blue = causal backbone, green = evidence, purple = restatement, red/pink = the two sides of a contradiction, orange = resolving holding.

AeroParts shipped a batch of turbine blades certified to spec^{a1}, and the blades were installed in the carrier's A-320 fleet^{a2}. The blades entered regular service across the fleet^{a3}. During service, seventeen blades failed in flight^{a4}. The regulator opened a safety probe^{a5}, and within a week issued an order grounding the fleet^{a6}; in effect, every affected aircraft was pulled from service^{a8}. Maintenance logs recorded abnormal vibration on the failed units^{a9}. The airline stated that no passengers or crew were harmed^{a7}. A separate wire report, however, claimed that three passengers died in one of the failures^{a10}. Early in the inquiry the airline argued that the failures were caused by pilot error^{a11}. The final accident report concluded that the root cause was a metallurgical defect in the AeroParts blades^{a12}, and on that basis found the pilots not at fault^{a13}. Because AeroParts had shipped the defective batch, the tribunal named AeroParts liable^{a14} and ordered it to pay damages^{a15}. The regulator also published a preliminary bulletin^{a16}.

Atomic units (self-contained, in order of appearance)

- a1. AeroParts shipped a batch of turbine blades certified to spec.
- a2. The turbine blades were installed in the carrier's A-320 fleet.
- a3. The turbine blades entered regular service across the fleet.
- a4. Seventeen turbine blades failed in flight during service.
- a5. The regulator opened a safety probe into the blade failures.
- a6. The regulator issued an order grounding the fleet.
- a7. The airline stated that no passengers or crew were harmed in the incidents.
- a8. Every affected aircraft was pulled from service.
- a9. Maintenance logs recorded abnormal vibration on the failed blades.
- a10. A wire report claimed that three passengers died in one of the failures.
- a11. The airline argued that the failures were caused by pilot error.
- a12. The final accident report concluded the root cause was a metallurgical defect in the AeroParts blades.
- a13. The accident report found the pilots not at fault.
- a14. The tribunal named AeroParts liable.
- a15. The tribunal ordered AeroParts to pay damages.
- a16. The regulator published a preliminary bulletin.

(Ids follow narration except that a8 — the restatement of a6 — is numbered out of order to keep the equivalence pair adjacent; a7 precedes a10 in the text.)

Mined relations (Layer B → Layer A, with strengths p)

Entailment / prerequisite / evidence (entail).

- $a1 \rightarrow a2$ (Cause-Effect, 0.90); $a2 \rightarrow a3$ (0.85); $a3 \rightarrow a4$ (Cause-Effect, 0.70); $a4 \rightarrow a5$ (0.80); $a5 \rightarrow a6$ (0.75).
- $a4 \rightarrow a7$ (Cause-Effect, 0.65) — failures are the events in which harm would or would not occur.
- $a9 \rightarrow a4$ (Evidence, 0.72) — vibration logs are evidence for the failures.
- $a13 \rightarrow a12$ (0.85) and $a12$ grounds $a13$ — the "not at fault" holding rests on the defect finding.
- $a1 \rightarrow a14$ (Evidence, 0.78); $a14 \rightarrow a15$ (0.70); $a5 \rightarrow a16$ (weak, 0.60).

Equivalence / restatement (equiv).

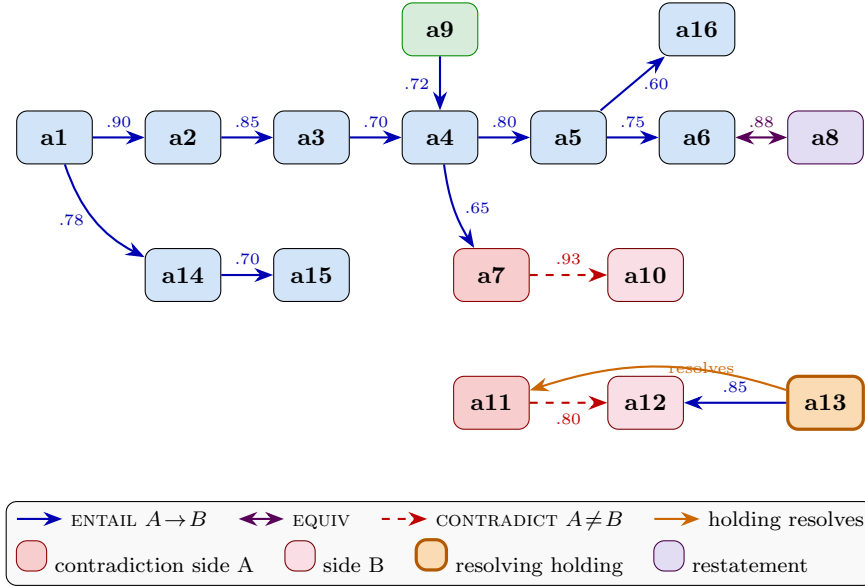
- $a6 \leftrightarrow a8$ (Restatement, 0.88) — "grounding order" $\equiv$ "aircraft pulled from service."

Contradiction (contradict).

- $a7 \neq a10$ (Contrast, 0.93) — *unresolved*: "no one harmed" vs. "three died." Both asserted as fact; nothing in the text adjudicates.
- $a11 \neq a12$ (Concession, 0.80 raw) — *resolved*: "pilot error" is superseded by the report's "metallurgical defect" finding, which the holding a13 adjudicates. This edge should be *discounted* (Section 2.3).

Relation graph

Solid blue = ENTAIL (labeled with p); double violet = EQUIV; dashed red = CONTRADICT; orange = the resolving holding a13. The unresolved contradiction $a7/a10$ is the coherence defect; $a11/a12$ is a tension the text resolves.



4 Relation confidence and uncertainty

Each p_{ij} becomes an edge/factor strength. Reuse FactReasoner's two backends (`--nli-method`): **logprobs** (from the generated label's token logprobs; fast, default) and **SIMBA-UQ** [SIMBA-UQ] (self-consistency across sampled judgments; backend-agnostic, better-calibrated). Two cautions carry extra weight here: (i) errors compound multiplicatively along a relation chain, so *calibration* (temperature scaling [Calib]) matters more than for single-atom factuality; and (ii) we need both the *type posterior* $P(\tau_{ij})$ and the *conditional strength* $P(a_j | a_i, \tau_{ij})$ — their product is the effective p that enters a factor. If the miner is unsure whether a pair is CONTRADICT vs. NONE, that uncertainty should *soften* the contradiction, not be thresholded away.

5 Model 1 — Undirected Markov Random Field (FactReasoner-style)

5.1 Construction (reuses the existing code path)

Each atom is a binary variable with a unary prior factor $[1 - \pi, \pi]$ ($\pi = 0.5$ for coherence-only; or the factuality FSS for a joint model). Each relation is a pairwise factor over $[src, trg]$, row-major

$(0, 0), (0, 1), (1, 0), (1, 1)$. FactReasoner ships *two* table variants (`assessor.py::edge_factor_values`); the choice is the key modeling decision for coherence.

Type	no-priors (atom $\leftrightarrow$ context default)	with-priors (recommended for atom $\leftrightarrow$ atom)
ENTAIL	$[p, p, 1-p, p]$	$[1-\pi_s, \pi_s, 1-p, p]$
CONTRADICT	$[p, p, p, 1-p]$	$[1-\pi_s, \pi_s, p, 1-p]$
EQUIV	$[p, 1-p, 1-p, p]$	$[p, 1-p, 1-p, p]$

The joint is $P(\mathbf{a}) = \frac{1}{Z} \prod_k f_k(\mathbf{a})$; inference is Merlin (WMB, i-bound 6, MAR task) returning posterior marginals $P(a_i)$; we also request Z (PR task) for Section 7.

Modeling finding: use the with-priors variant for coherence

The no-priors ENTAIL table $[p, p, 1-p, p]$ penalizes the *source* being true when the target is uncertain (the $(1, 0)$ cell is $1-p$). That is desirable for atom $\leftrightarrow$ context factuality — an atom with no supporting context *should* be pulled down — but wrong for atom $\leftrightarrow$ atom coherence, where a well-connected *premise* must not be penalized merely for being a premise. Exact inference confirms it: under the no-priors tables the backbone root `a1` collapses to $P(a1) = 0.19$ and mean support is 0.46 even after *removing* all contradictions. The with-priors tables decouple the source’s own probability from the implication and give the intended behaviour (Table 1). **We therefore adopt the with-priors tables ($\pi_s = 0.5$) for atom $\leftrightarrow$ atom edges.**

5.2 Exact marginals on the example

Table 1 gives exact posteriors (brute force over 2^{16} worlds) for three model variants: **base** (both contradictions active), **concession-discounted** (the resolved `a11`≠`a12` softened $0.80 \rightarrow 0.55$), and a hypothetical **coherent rewrite** (all contradictions removed).

	a1	a2	a3	a4	a5	a6	a7	a8
base	.503	.706	.761	.743	.723	.681	.612	.637
	a9	a10	a11	a12	a13	a14	a15	a16
base	.531	.237	.441	.528	.441	.641	.628	.572

Table 1: Exact MRF posterior marginals $P(a_i=1)$ for the base model (with-priors tables, $\pi=0.5$). The backbone atoms `a2`–`a6` rise to 0.68–0.76 (well-supported); the loser of the strong casualty contradiction, **`a10`**, is driven to 0.24; the softer blame contradiction leaves `a11`/`a12` near 0.44/0.53.

The **MAP configuration** (most probable joint world) is the interpretable ”best reading”: it sets every backbone atom true, and resolves both contradictions decisively — **`a7` = 1, `a10` = 0** (accept ”no casualties”, reject ”three died”) and **`a11` = 0, `a12` = 1** (reject the retracted pilot-error blame, keep the adjudicated defect finding), with $P(\mathbf{a}^*) = 4.49 \times 10^{-3}$.

6 Model 2 — Markov Logic Network

An MLN [MLN, Markov Logic] is a set of weighted first-order formulas (F_k, w_k) ; grounding over the atoms yields a Markov network with $P(\mathbf{a}) = \frac{1}{Z} \exp(\sum_k w_k n_k(\mathbf{a}))$, where n_k counts true groundings of F_k . It was used for textual entailment by (author?) [RTE-MLN]. Its advantage over the pairwise

MRF is *expressivity*: relations become first-order rules, and we can write rules that pairwise factors cannot.

6.1 Rule schema for coherence

$$\begin{aligned}
(\text{entail}) \quad w_e \cdot p_{ij} : & \text{Entail}(i, j) \wedge a_i \Rightarrow a_j \\
(\text{contradict}) \quad w_c \cdot p_{ij} : & \text{Contradict}(i, j) \wedge a_i \Rightarrow \neg a_j \\
(\text{equiv}) \quad w_q \cdot p_{ij} : & \text{Equiv}(i, j) \Rightarrow (a_i \Leftrightarrow a_j) \\
(\text{transitivity}) \quad w_t : & \text{Entail}(i, j) \wedge \text{Entail}(j, k) \Rightarrow \text{Entail}(i, k) \\
(\text{concession cancels}) \quad w_r : & \text{Resolves}(h, i) \wedge a_h \Rightarrow \neg \text{Penalize}(\text{Contradict}(i, j))
\end{aligned}$$

The last two are the point: **transitivity** lets a weak two-hop chain (a3→a4→a7) contribute joint evidence a pairwise model treats as independent; the **concession-cancels** rule fires on the resolving holding a13 and *removes* the a11≠a12 penalty when a13 holds — a first-class treatment of ”resolved tension” that the MRF can only approximate by hand-softening p (the $0.80 \rightarrow 0.55$ discount of Section 5). Higher-arity rules (”an atom contradicted by two independent later atoms is doubly penalized”) are equally expressible.

6.2 Inference and scoring hooks

MAP inference (MaxWalkSAT) gives the most-consistent world and its violated-weight energy; marginal inference (MC-SAT / lifted BP) gives per-atom probabilities analogous to the MRF. A coherence score is the MAP energy $-\sum_k w_k n_k^{\text{viol}}(\mathbf{a}^*)$ (squashed to $[0, 1]$), or $P(R)$ for a reified rule (Section 7).

6.3 MRF vs. MLN — when to use which

	MRF (Model 1)	MLN (Model 2)
Weights	p enters factor directly; <i>no learning</i>	log-space w ; needs learning/tuning
Cycles / bidir.	native (undirected)	native (undirected)
Expressivity	pairwise only	full first-order (transitivity, concession-cancel, n -ary)
Inference cost	sub-second (Merlin WMB), shipped	#P-hard; MC-SAT / MaxWalkSAT, new dep.
Concession	hand-softened p	rule that <i>cancels</i> the penalty
Recommendation	primary (max reuse, sound)	research branch for rules over relations

The MLN is a strict generalization: with only the pairwise (entail/contradict/equiv) rules and appropriate weights it reproduces the MRF. Adopt the MRF as the default and reach for the MLN exactly when we need rules *over* relations.

7 Scoring the coherence of the whole response

We work every applicable readout on the example (base MRF, with-priors). They are not mutually exclusive; the recommendation is to report the reified-node score with the Z -ratio and mean-support as companions.

S1 — Joint / geometric-mean product (BN-style, Section 6.1 of the plan). The raw joint $\prod$ of relation strengths decays with length; use the length-normalized geometric mean. Over the 11 ENTAIL links the product is 4.19×10^{-2} and the geometric mean is **0.749**. *Pro*: intuitive. *Con*: ignores contradictions unless folded in; needs a DAG for a true joint (cycles here forbid it).

S2 — Aggregate of posterior marginals (MRF). From Table 1: **mean marginal support** = 0.587 (base) vs. 0.601 (concession-discounted) vs. 0.620 (coherent rewrite) — monotone in coherence, as desired. **Average normalized entropy** (FactReasoner’s $\mathcal{E}$) = 0.925 (base), higher than the coherent rewrite: contradictions push marginals toward 0.5, raising entropy. **Atoms dragged below their prior**: 3 (base), a crisp diagnostic count (a10, a11, a13 — the losers/attachments of the two conflicts).

S3 — Partition-function ratio (Section 6.3). $\text{LCS} = Z_{\text{full}}/Z_{\text{indep}}$, or its log. Here $\log Z_{\text{base}} = -9.75$ vs. $\log Z_{\text{coherent}} = -8.25$: removing the contradictions raises Z by ≈ 1.5 nats — the structure is ”more jointly satisfiable.” This is the most sensitive global signal to contradictions (needs Merlin PR task) but is not itself in $[0, 1]$ without a reference.

S4 — Reified coherence node R (Section 6.4) — RECOMMENDED. Add one binary R with a factor favouring $R=1$ when ENTAIL/EQUIV edges are satisfied and CONTRADICT edges inactive; report $\text{LCS} = P(R=1)$. A single calibrated $[0, 1]$ number, length-robust, no DAG needed, one extra factor in the existing MarkovNetwork. This is the clean bridge from per-atom marginals to an argument-level score.

S5 — MAP energy / PSL satisfaction (Section 6.5). MRF/MLN: report the MAP world and its (normalized) probability $P(\mathbf{a}^*) = 4.49 \times 10^{-3}$, or the count of violated relations at the MAP world (1: the unresolved a7/a10). PSL [PSL] gives $1 - \text{mean distance-to-satisfaction}$ directly in $[0, 1]$.

S6 — α/β support-vs-contradiction knob (Section 6.6). $\text{LCS} = \sigma((\alpha \sum_{\text{ent}} p - \beta \sum_{\text{con}} p)/|E|)$ with $\beta > \alpha$ penalizing contradictions harder. With support mass $S = 9.18$, contradiction mass $C = 1.73$: $(\alpha, \beta) = (1, 1) \rightarrow 0.630$; $(1, 2) \rightarrow 0.601$; $(1, 3) \rightarrow 0.571$. A cheap post-hoc layer to calibrate on human ratings, usable atop any core score.

	Readout	base	concession-disc.	coherent rewrite
S1	geo-mean of ENTAIL p	0.749	0.749	0.749
S2	mean marginal support	0.587	0.601	0.620
S2	avg. normalized entropy $\downarrow$	0.925	—	lower
S3	$\log Z_{\text{full}}$	−9.75	−9.42	−8.25
S5	MAP prob $P(\mathbf{a}^*)$	4.5×10^{-3}	—	—
S6	$\sigma, (\alpha, \beta) = (1, 2)$	0.601	—	—

Table 2: All applicable scoring readouts on the worked example. S1 (relation-only) is blind to contradictions — it cannot tell the three variants apart — which is exactly why S2/S3/S4 (inference-based) are preferred: they *decrease* for the contradiction-laden base and *recover* as tensions are resolved or removed.

7.1 Scoring-behavior figure

Figure 1 plots the two inference-based global scores (mean support, and $\log Z$ rescaled) across the three variants, against the contradiction-blind S1 baseline.

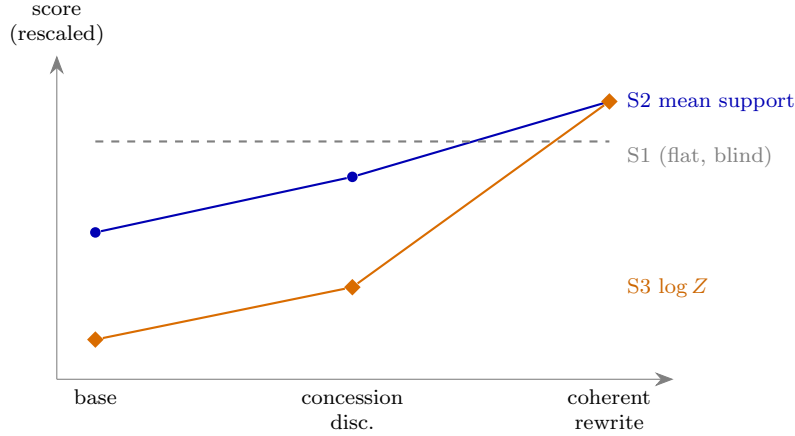


Figure 1: Global coherence scores across the three variants (higher = more coherent). The inference-based scores S2 (mean marginal support) and S3 ($\log Z$) rise as contradictions are resolved/removed; the relation-only baseline S1 is flat because it never sees the contradictions. Values are those of Table 2, each linearly rescaled to a common axis for shape comparison.

8 Synthesis and recommendations

1. **Taxonomy.** Adopt the two-layer scheme: mine an interpretable PDTB/RST *discourse* sense, map it to an ENTAIL/CONTRADICT/EQUIV *coupling* with a graded strength. Treat Concession / adjudicated holdings as *resolved* tensions, not defects.
2. **Primary model: the pairwise MRF** (Model 1), reusing MarkovNetwork + Merlin, **with the *with-priors* factor tables** for atom $\leftrightarrow$ atom edges (the no-priors tables mis-penalize premises — Section 5). This is a small delta on shipped code: add `link="atom_atom"` edges and one reified factor.
3. **Research branch: the MLN** (Model 2) when we need rules *over* relations — transitivity of entailment and, especially, a concession-cancels-penalty rule that the MRF can only approximate by hand-softening p .
4. **Primary score: the reified coherence node $P(R=1)$ (S4)**, reported with **mean marginal support** (S2) and the **$\log Z$ ratio** (S3) as companions. Avoid relation-only products (S1) as the headline number — they are blind to contradictions (Table 2). Keep the α/β knob (S6) as a calibration layer.
5. **Evaluation.** Unit-test the miner and the scores on the five ideation PDFs plus this example; stress order-sensitivity on *Renda* S vs. K; correlate against human coherence ratings, benchmarking DiscoScore [DiscoScore] and a G-Eval [G-Eval] LLM-judge; ablate logprobs vs. SIMBA-UQ and with/without calibration.

Reproducibility. All marginals, Z , and MAP values in Sections 5–7 are exact (brute force over 2^{16} worlds) for the model of Section 3 using the with-priors factor tables of `assessor.py`; they are intended to be regenerated by a small reference solver as the design is prototyped against Merlin.

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